March 15, 2025

Faisal D'Souza, NCO
Office of Science and Technology Policy
Executive Office of the President
2415 Eisenhower Avenue Alexandria, VA 22314

Submitted by email to [redacted]

Re: Request for Information (RFI) on the Development of an Artificial Intelligence Action Plan ("Plan").

Introduction

EdgeConneX supports the Office of Science and Technology Policy's (OSTP) efforts to develop an actionable plan to support the United States' leadership in the development and deployment of artificial intelligence (AI) technologies. We are grateful for the opportunity to contribute to this RFI and hope to be a close partner of the Trump Administration as it seeks to address the myriad policy issues which serve as limiting factors for the advancement of the AI economy.

In a very real sense, data centers are launchpads. We don't create cloud services, AI models, or streaming content. But we enable, empower, and accelerate these, and, by extension, virtually every other major industry in the world. From agriculture to transportation, healthcare to technology, manufacturing to logistics, they all rely on today's digital infrastructure platforms, which originate and terminate in the world's data centers.

Our recommendations address policy challenges which, in our view, are the largest limiting factors for the deployment of data centers - the center of the digital economy. Data centers are where the energy industry meets computing capabilities to connect consumers and enterprises to online services which advance the modern economy. These facilities are hubs of physically interconnected industries, and as such there are several federal policy issues that could impact our work.

About EdgeConneX

Founded in 2009, EdgeConneX is focused on driving innovation and helping our customers define and deliver their own unique vision for the Edge, at any scale, in any market worldwide, for any requirement. Today, with headquarters in Northern Virginia, Singapore, and Amsterdam, we are building tomorrow's data center infrastructure.

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Delivering innovative and proximate data center solutions ranging from 40kW to 100MW or more, we work closely with our customers to provide the scalable capacity, power, and connectivity they need to meet the growing demands of their business and their end users. In turn, our customers' content, connectivity, cloud services or any data requirement can be securely delivered with enhanced performance and lower latency to any device, anywhere.

Since late 2013, EdgeConneX has over 80 data centers built and in development, including Edge Data Centers® and a growing number of regional and hyperscale solutions across North America, Europe, Asia Pacific, and South America, creating a new Edge of the Internet.

EdgeConneX continues to move the Internet to where it is most needed – into local markets across the country and around the globe. We recognize that demand for data, content, cloud services, and ubiquitous computing is growing each day and that quality delivery of this content requires a rapid response, global expertise, and a focus on delivering sustainable data center solutions worldwide.

Policy Recommendations

1.) Collaborate with state and local governments to reduce permitting burdens by setting standards for processing applications related to the construction of data centers and related energy infrastructure.
2.) Utilize federal programs to incentivize and expand the growth of our nation's skilled trade labor workforce, including specialized construction trades.
3.) Liaise with state governments to ensure that public and private strategic investments in AI infrastructure can be deployed in a timely manner, and in a way that benefits American communities across the country.
4.) Direct federal agencies to expedite transmission line approvals to improve the capacity and resiliency of our nation's energy transmission grid.
5.) Continue making strategic investments in federal R&D efforts that can accelerate the adoption of new power generation capabilities.
6.) In international markets of strategic importance to the U.S., we must ensure that U.S. companies can remain competitive – from land acquisition through project construction – to protect foreign market access so that American companies benefit from technology transitions at home and abroad. The US Government should work to protect US vendors from discrimination abroad; it is an economic and national security imperative that US companies are supported, and can remain competitive against foreign competitors in major markets.

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Powering and Empowering AI Infrastructure Deployment

Data centers are significant consumers of electrical power, housing cooling units, servers, networking and storage equipment, and other devices, all of which require 24x7 availability and operation. Much of this power has traditionally been generated using fossil fuels that contribute to significant carbon emission, and shifting to more sustainable energy can be challenging.

A key resource required by all data centers is electrical power, and as the reach and adoption of the Internet and digital business transformation have grown, data center electricity usage has grown significantly even as overall efficiency has improved. According to recent analysis from Boston Consulting Group, data centers already make up roughly 2.5 percent of total U.S. electricity demand, but exploding demand for AI could drive that to 7.5 percent by 2030.[1] Further, there are indications that improvements in Power Usage Efficiency (PUE) have slowed, which may accelerate energy consumption going forward. This is due in large part to older, less efficient data centers remaining in service, making further reductions toward a 1.0 industry-wide PUE challenging even with newer data centers coming online.

We recommend directing federal agencies to expedite transmission line approvals to improve the capacity and resiliency of our nation's energy transmission grid. EdgeConneX selects locations for data center projects based on the needs of its customers, and location selection is heavily dependent on the availability of power supply. To provide high-latency services amidst increasingly high demands for data throughput, data centers need to be constructed close to major populations hubs, however, those markets are the first to become energy-constrained – both in supply, and in transmission throughput.

We recommend that Congress and the Administration continue making strategic investments in federal R&D efforts that can accelerate the adoption of new power generation capabilities. To solve for power constraints in target US markets, ECX has started building hydrogen-ready natural gas power plants – collocated with our data center builds - to meet the immediate needs of our customers as a stop-gap solution to longer-term, renewable energy supply.

It is our strategic vision that, in the future, renewable energy solutions can supply 24/7 power generation for our data center builds. Natural gas engines and turbines are currently available to offset local gaps in power supply, and they are also convertible facilities which may be converted to cleaner hydrogen fuel when available.[2]

[1] https://www.linkedin.com/pulse/impact-genai-electricity-how-fueling-data-center-boom-vivian-lee/
[2] https://www.energy.gov/articles/doe-releases-new-report-evaluating-increase-electricity-demand-data-centers

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Federal R&D efforts should continue to revolutionize energy solutions, while also working to incorporate best practices into the private sector. In our view, small modular nuclear reactors are a possible solution, albeit untested in commercial applications, and their integration into existing grid infrastructure will take at least a decade until a fully viable solution to support data center energy demands. Photovoltaic and wind energy, paired with battery storage, are high-cost and not currently scalable to meet the large, immediate demands for energy supply.

ECX is putting renewables solutions to work in the US with a 1 Gigawatt campus planned for deployment in a major market where other power generation solutions are still several years away. However, these are high-cost projects, and the scalability of these renewable energy solutions are not applicable for all deployments.

Making AI Local

Over the past decade, data centers have moved physically closer to both enterprises and the American consumer as the demand for data to be pushed to the edge has caused data centers and interconnection facilities to move closer to locations of high demand. However, the placement of these facilities is often a complex equation which depends on availability of energy supply, transmission grid efficiency, and telecommunications access, in addition to a broad range of on-the-ground factors.

With over 30 deployments within the U.S., we are responsive to a growing number of constituencies, many with diverse social and regulatory priorities and processes. Our customers demand practical, sustainable options for energy, water, land, and operations – and oftentimes, so do our partners in state and local governments. As a data center provider, we are accountable not only to our customers, but also directly to American communities for the construction of large-scale, longitudinal construction projects, each with their own unique footprint and logistical and community-based challenges.

We recommend that the Administration work closely with state and local governments to create more ubiquitous standards for the construction and deployment of data centers. State and local governments are a prime constituency of data center providers; unlike software developers and content creators which run on the infrastructure that we build, we are the primary interface with local governments as we seek to acquire land, permit facilities, and hire skilled labor across diverse localities. Given the size and scope of these construction projects, unnecessary delays can have a burdensome impact on the cost and speed at which new facilities can be built, which in turn will have a direct impact on the deployment of new AI enabled services in American communities.

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Building Generational Employment Opportunities

Data centers – even the smaller-scale Edge Data Centers (EDCs) that our company is known for – are extraordinarily large construction projects which rely upon a resilient, skilled labor force in local communities across the United States. After acquiring land and receiving the requisite permits and licenses to begin a new build – which may alone take months or years in some cases – the real work begins to construct these facilities which have highly specialized requirements to meet the energy and efficiency demands of our customers while ensuring stable, secure operations of a facility.

We recommend utilizing federal programs that can incentivize and expand the growth of our nation's skilled trade labor workforce, including specialized construction trades. These construction projects are typically multi-year efforts and require a broad range of skilled laborers. These individual projects create thousands of jobs, either directly or indirectly, through the hiring of construction contractors which work locally to source and retain laborers that can commit to long-term workflows.

Across existing EdgeConneX projects sites within the US, labor shortages have contributed to an average labor premium increase of 20% as our company often has to pay advanced retainers for skilled laborers over long contract terms, pulling those laborers away from other projects.

Large-scale data centers can require thousands of laborers over a multi-year construction duration. This often challenges local communities to deliver other public and private infrastructure projects in a timely manner. In our industry, and across other large-scale land developments, the supply of skilled labor falls far below the demand, driving up costs and delaying the timeframe for delivering new AI-enabled products and services.

We recommend pursuing a nationwide push to incentivize scaling workforce development programs within communities where supply is limited. Federally-supported training programs, like technical education and apprenticeship initiatives, should more closely partner with industries that have a high demand nexus – like data center providers – which can help provide training resources, while also developing specialized expertise in AI-related subspecialities which will be necessary for the completion of complex projects across the US for the foreseeable future.

Conclusion

Thank you for the opportunity to contribute to this Artificial Intelligence Action Plan.

EdgeConneX supports the Office of Science and Technology Policy's (OSTP) efforts to develop an actionable plan to support the United States' leadership in the development and

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deployment of artificial intelligence (AI) technologies. We are encouraged by the Trump Administration's continued interest in championing AI policy, and we hope to be a close partner as you seek to address the myriad policy issues which serve as limiting factors for the advancement of the AI economy.

Our business has a direct and immediate impact on communities in the US and across the world. We are building the Internet's edge, ensuring that enterprise customers and individual consumers can connect to meet their data needs. Over the last decade, that edge has changed drastically, and to adequately support the future AI economy, we will need to continue to deploy EDCs quickly, and in a manner which supports the sustainability and resiliency of our nation's energy grid. We also must seize the opportunity to benefit the American labor force, expanding opportunities for skilled laborers around the country to benefit from the rapid growth of data centers which will be necessary to facilitate the broader adoption of AI technologies.

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